ANSYS Apache

GPS Tool Tcl Command Reference

Version 1.0

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DATA ENTRY COMMANDS

The below commands are used to input various data and pointers required for GPS.

Command	Options	Allowed Arguments	Description	Example
data read	defs	<def> { {<filename [block|top] [import_all]>}... }	Provides path to def files. "import_all" means import all DEF elements like STD cell , NETS , SPECIALNETS etc. Without "import_all" , only flooplan will be imported	data read defs { { /home/sankar/block1.def block import_all } { /home/sankar/block2.def block } { /home/sankar/top.def top import_all } }
	lefs	<Path to LEF Files>	Provide path to LEF files . Technology lef must come first	data read lefs { /home/sankar/tech.lef /home/sankar/std_cell.lef /home/sankar/mem.lef }
	ploc	Pointers to PLOC Files	Provide path to ploc files which contains voltage source locations	data read ploc /home/sankar/my.ploc
	switch_model	Pointers to Switch Model File	Provide path to Switch Model Files which contain Electrical Model of Power Gates used in RedHawk analysis	data read switch_model /home/sankar/switch.model
config set	upf_files	Pointers to Synopsys Unified File Format (UPF) file	Users can pass UPF files for RH PS Engine based Power Calculation in GPS	config set upf_files { /home/sankar/1.upf /home/sankar/2.upf }
	cpf_files	Pointers to Common File Format (CPF) file	Users can pass CPF files for RH PS Engine based Power Calculation in GPS.	config set cpf_files { /home/sankar/1.cpf /home/sankar/2.cpf }
	ploc/ploc_files	Pointers to Ploc file	Users can provide pointers to Pad files which contain Voltage Source Locations.	config set ploc/ploc_files { /home/sankar/my.ploc }
	sdc_files	Pointers to SDC files	Users can provide pointers to SDC files which will be used in RH PS for timing and power computation.	config set sdc_files { /home/sankar/my.sdc }
	verilog_files	Path to Verilog Files	Provide the path to Verilog files for RH PS based Power Computation.	config set verilog_files { /home/sankar/1.v /home/sankar/2.v }
	tech	Path to Apache Technology file	Provide the path to Apache Tech File which will be passed to RedHawk for Resistance computation and EM analysis.	config set tech { /home/sankar/apache.tech }
	libs	Pointers to Synopsys Liberty Files	Users can provide pointers to Liberty files for RH PS and RedHawk analysis.	config set libs { /home/sankar/1.lib /home/sankar/2.lib }

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CONFIGURATION COMMANDS

The below commands are used for setting basic configuration related to GUI , DRC etc.

Command	Options	Allowed Arguments	Description	Example

config set	additional_gsr_settings	<Any GSR setting>	Users can pass any extra gsr setting for RedHawk Analysis through this keyword	config set additional_gsr_settings {
				REPORT_REDUCTION 2
				ENABLE_AUTO_EM 1
				VCD_FILE {
				…….
				}
				}

	frequency	<Freq value in Hz>	Users can pass FREQUENCY setting to RedHawk gsr through this setting. Default is 0.	config set frequency 500e6
	gui_canvas_refresh	<0/1>	Users can turn off/on canvas refresh by setting it to 0/1 respectively. Setting it to 0 during extensive GUI operations saves runtime. Default is 1.	config set gui_canvas_refresh 0
	gui_incr_canvas_draw	<0/1>	Users can turn off/on incremental canvas redraw using this keyword. Setting it to 0 will increase runtime for GUI operations. Default is 1.	config set gui_incr_canvas_draw 0
	gui_parallel_canvas_draw	<0/1>	Default 1. Recommended for fast GUI executions.	config set gui_parallel_canvas_draw 0
	min_spacing	<floating_point_value>	Global Minimum Spacing constraing for avoiding shorts.	config set min_spacing 0.1' sets a minimum distance of 0.1 um during grid creation between grid elements for avoiding shorts
	multithreaded_def_read	<0/1>	Default 1. Recommended for fast DEF reading. If set to 1 , reads DEF in multithread mode	config set multithreaded_def_read 0' turns off multi threaded def reading
	offset_from_blockages	<floating_point_value>	Default 0 um. This represents the distance at which grid creation stops from blockages.	config set offset_from_blockages 2.5
	offset_from_exclude_scopes	<floating_point_value>	Default 0 um. This represents the distance at which grid creation stops from scopes or blocks which are excluded from grid creation.	config set offset_from_excluded_scopes 3.2
	rh_apache_root	Pointer to RedHawk binary path	Users can set the path for RedHawk APACHEROOT from where RH binary will be invoked for analysis.	config set rh_apache_root { /apps/apache/14.2.1 }
	rh_post_analysis_tcl	Pointer to Extra TCL file for RH post analysis	Users can add additional commands in a TCL file for RH analysis. The commands will be added at the end of the TCL file that GPS creates for RedHawk.	config set rh_post_analysis_tcl { /home/sankar/my.tcl }
	short_check	<0/1>	Default 1. Short Check is always enabled. Can be set to 0 to avoid shorts during grid creation.	config set short_check 0
	tcl_auto_complete	<0/1>	Default 1. Auto completes the TCL command if command is entered partially in the GUI. Please use tab key for auto completion.	config set tcl_auto_complete 0

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FLOORPLAN COMMANDS

The below commands are used to perform various floorplan operations.

Command	Options	Allowed Arguments	Description	Example
db create	floorplan	-name <> -offsets { T L B R } -size { <width> <height> } -origin { <x> <y>}	Used to create a floorplan from scratch. -name switch is used to mention the name of design -offsets switch is used to define the Top Left Bottom and Right Offsets for I/O region and Core Region Differentation -size switch is used to define width and height of floorplan -origin switch is used to define the origin of design floorplan	db create floorplan -name Top -offsets { 30 40 30 40 } -size { 2200 2000 } -origin { 25 25 }
	inst	inst <parent cell name> -name <name> -master <name> -loc {<x> <y>} -orient <N | W| S | E | FN | FW | FE | FS> -logical_conn { {pin_name net_name} ..}	Used to insert an instance at a location in design <parent cell name> defines the block/cell in which the instance needs to be inserted -name switch is used to mention name of instance -loc switch is used to specify x,y location of instance -orient switch is used to mention orientation of instance. -master switch is used to mention master name of the instance	db create inst /work/Top -name inst1 -loc {-1041.76 2589.11} -master /work/invx1 -orient N
	net	<parent cell name> -name <name> -type <p|g> -volt <val> -logical_conn { <inst1> <pin1>} <inst2> <pin2> .. }	Used to add a new net in design <parent_cell_name> defines the block/cell in which the net needs to be inserted -name switch defines the new net name -type <p|g> switch mentions if it is power(p) or ground(g) net -volt <value> switch mentions the voltage value for the net -logical_conn switch is used to add logical connectivity of the net to design elements	db create net /work/TOP -name VDD -type p -volt 1 -logical_conn { { switch_* VDDC } }
	placement_blockage	<parent_cell> -name <name> -rect {<llx> <lly> <urx> <ury>}	Used to add a new placement blockage in design <parent_cell> mentions the block/cell under which the blockage is created. -name mentions the name of the blockage which is created -rect mentions the bounding bbox of the placement blockage	db create placement_blockage /work/TOP -name PB1 -rect { 456.426 298.346 750.4 475.84 }
	region	<parentcell> - name <name> - rect {<llx> <lly> <urx> <ury>} | -rectilinear { {<x1> <y1>} {<x2> <y2>} {<x3> <y3>} ..}	Used to add a new region in the design <parent_cell> mentions the block/cell under which the region is created. -name mentions the name of the region which is created -rect is used to mention the bbox of the rectangular region -rectlinear is used to mention the coordinates if rectilinear region is needed.	db create region {/work/TOP} -name DSP -rect { 275 275 1485 1425 }
	routing_blockage	<parent_cell> - name <name> - layers {<layer1> <layer2> ..} -rect {<llx> <lly> <urx> <ury>} | - rectilinear { {<x1> <y1>} {<x2> <y2>} {<x3> <y3>} ..}	Used to add a new routing blockage in the design <parent_cell> mentions the block/cell under which the blockage is created. -name mentions the name of the blockage -layers is used to mention layers for which blockage is created -rect is used to mention the bbox of the rectangular blockage -rectlinear is used to mention the coordinates if rectilinear blockage is needed	db create routing_blockage /work/TOP -name RB1 -rect { 1950 225 2550 465 } -layers { m6 m5 m4 m3 m2 m1 via5 via4 via3 via2 via1 }

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MESH COMMANDS

The below commands are used to perform various mesh related operations like route mesh , via mesh , switch/ploc mesh,etc.

Command	Options	Allowed Arguments	Description	Example
db create/edit	pin_tap_spec/ <pin_tap_spec_name> Provide pin_tap_spec keyword for db create command. For db edit , provide <pin_tap_spec_name> name instead of pin_tap_spec keyword. See example on right side for more details.	<parent_cell> -name <name> -route_layer <name> -pin_layers {<layer1> <layer2> ..} -all_pin_layers -nets {<net1> <net2> ..} -all_nets -insts {<inst1> <inst2> ..} -all_insts -all_blocksall hardips and blocks-via_models {<name1> <name2> ..} -via_rules {<name1> <name2> ..} -via_occupancy_percentage {<XVal> <YVal>} -via_occupancy {<XVal> <YVal>} -via_num_cuts {<NX> <NY>} -vias_between_parallel_metals 0|1-parallel_metals_via_spec {<length> <spacing>} -restrict_via_cuts_to_overlap <0|1>	Used to create a pin tapping spec for Hard Macro Pin Tapping or via insertion on hard macro pins <parent_cell> mentions the block/cell under which the spec is create.Used only for db create command. Not used in db edit command. -name mentions the spec name created for pin tapping -name mentions the spec name created for pin tapping -pin_layers Is used to mention which pin layers must be tapped on hard IPs -all_pin_layers means all pin layers in hardIP will be used for macro pin tapping -nets is used to mention nets for which vias needs to be inserted -all_nets means all nets need via insertion -insts is used to mention specific IPs for hard macro pin tapping -via_models is used to mention specific via models to be used for via insertion -via_rules is used to mention specific via rules to be used for via insertion -via_occupany_percentage is used to mention the percentage occupancy of via -via_occupancy is used to mention absolute x and y values for via occupany -via_num_cuts is used to mention number of cuts in x and y direction -vias_between_parallel_metals 0/1 is used to control if vias need to inserted between parallel metal layers -parallel_metals_via_spec { <length> <spacing>} is used to mention the length and spacing of vias between parallel metal layers -restrict_via_cuts_overlap is used to limit the via cuts to the overlap area between metals	db create pin_tap_spec /work/ap_top -name PT2 - route_layer metal5 -all_pin_layers -all_nets -insts { /work/ap_top/inst1 } -vias_between_parall el_metals 1 -parallel_metals_via_spec { 0 0 } -via_occupancy_percentage { 100 100 } db edit /work/TOP/pin_tapping:PT1 -name PT1 -route_layer M6 -all_pin_layers -nets { VDD_TOP } -via_rules {VIA1_0 } -restrict_via_cuts_to_overlap 1 -vias_between_parallel_metals 1 -parallel_metals_via_spec { 0 0 } -via_occupancy_percentage { 100 100 }
	ploc_mesh/ <ploc_mesh_spec_name> Provide ploc_mesh_spec keyword for db create command. For db edit , provide <ploc_mesh_spec_name> name instead of ploc_mesh_spec keyword. See example on right side for more details.	<parent_cell> -name <name> -layer <name> -spacing {<x> <y>} -offsets {<top> <left> <bottom> <right>} -spec_boundary entire core| {bbox { x1 y1 x2 y2 } } |select_region exp { }	Used to create plocs or pads in the design <parent_cell> mentions the block/cell under which the spec is created.Used only for db create command. Not used in db edit command. -name mentions the spec name created for ploc mesh -layer mentions layer used for inserting pads -spacing mentions the x and y pitch between pads -offsets mentions the top left bottom and right offsets used for pad placement -spec_boundary is used to mention areas where route mesh will be created. If 'entire core' option is used , then all the core area will be used by route mesh spec. 'bbox' option can be used to create route mesh only within a rectangular region. '-select_region_exp' is used to provide an expression of scope or design blocks to determine where the grid must be created. Format is -select_region_exp{+block1 -block2 - ... } . '+' means inclusion and '-' means exclusion of a block/region when mesh is created.	db create ploc_mesh /work/Top -name Pad1 -layer MET5 -spacing { 250 200 } -offsets { 0 5 0 0 } -spec_boundary {entire_core } db edit /work/TOP/pm:Pad1 -name Pad1 -layer M6 -spacing { 101 200 } -offsets { 0 0 0 0 } -spec_boundary {entire_core }

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MESH COMMANDS

db create/edit	ring/ <ring_spec_name> Provide ring_spec keyword for db create command. For db edit , provide <ring_spec_name> name instead of ring_spec keyword. See example on right side for more details.	<parent_cell> -name <name> -nets {<net1> <net2> ...} -top {layer width spacing offset} -bot {layer width spacing offset} -left {layer width spacing offset} -right {layer width spacing offset} -drop_via_layers {<upper> <lower>} -bbox {<llx> <lly> <urx> <ury>}	Used to create a ring in the design <parent_cell> mentions the block/cell under which the spec is created.Used only for db create command. Not used in db edit command. -name mentions the spec name used for creating the ring -nets mentions the nets which will be there in the ring -left { layer width spacing offset } is used to mention the layer , width , spacing and offset used for the left portion of the ring -top { layer width spacing offset } is used to mention the layer , width , spacing and offset used for the top portion of the ring -right { layer width spacing offset } is used to mention the layer , width , spacing and offset used for the right portion of the ring -bottom { layer width spacing offset } is used to mention the layer , width , spacing and offset used for the bottom portion of the ring	db create ring /work/TOP -name RING1 -nets { VDD } -top { ALCAP 10 0 -1 } -left { MET6 10 0 -1 } -bot { ALCAP 10 0 -1 } -right { MET6 10 0 -1 } db edit /work/TOP/ring:RING1 -name RING1 -nets { VDD_TOP } -top { M1 1 0 -10 } -left { M1 1 0 -10 } -bot { M1 1 0 -10 } -right { M1 1 0 -10 } INFO(RING-000): In ring RING1, number of wires created: 4, number of vias created: 0

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MESH COMMANDS

db create/edit	route_mesh/ <route_mesh_spec_name> Provide route_mesh_spec keyword for db create command. For db edit , provide <route_mesh_spec_name> name instead of route_mesh_spec keyword. See example on right side for more details.	<parent_cell> -name <name> -dir <direction> -layer <name> -nets {<net1> <net2> ...} -width <val> -spacing <val> -pitch <val> -drop_via_layers {<upper> <lower>} -offsets {<top> <left> <right> <bot>} -spec_boundary entire core| {bbox { x1 y1 x2 y2 } exclude_rects { {a1 b1 a2 b2 } { a11 b11 a22 b22 } ... } } |select_region exp { } -via_models {<name1> <name2> ..} -via_rules {<name1> <name2> ..} -via_occupancy_percentage {<XVal> <YVal>} -via_occupancy {<XVal> <YVal>} -via_num_cuts {<NX> <NY>} -vias_between_parallel_metals 0|1-parallel_metals_via_spec {<length> <spacing>} -restrict_via_cuts_to_overlap <0|1>-short_check <1|0> -min_spacing <val> -offset_from_blockages <val> -offset_from_exclude_scopes <val> -staples {<length> <spacing>} -hide_spec <1|0>	Used to create a spec for route mesh on a specific layer <parent_cell> mentions the block/cell under which the mesh spec is create.Used only for db create command. Not used in db edit command. -name mentions the spec name created for route mesh -direction <h|v> is used to mention whether route mesh needs to be created in horizontal or vertical direction. -layer Is used to mention layer used for routing -nets is used to mention nets for which routes need to be created -width means width of strap in mesh spec -spacing means spacing between straps in mesh spec -pitch means group pitch between group of straps in mesh spec -offsets is used to mention top , left , bottom and right offsets for starting grid creation. -spec_boundary is used to mention areas where route mesh will be created. If 'entire core' option is used , then all the core area will be used by route mesh spec. 'bbox' option can be used to create route mesh only within a rectangular region. 'exclude_rects' option is used to provide single/multiple bbox where mesh creation needs to be avoided. '-select_region_exp' is used to provide an expression of scope or design blocks to determine where the grid must be created. Format is -select_region_exp{+block1 -block2 - ... } . '+' means inclusion and '-' means exclusion of a block/region when mesh is created. -drop_via_layers is used to mention upper and lower layers to which vias will be dropped. -via_models is used to mention specific via models to be used for via insertion -via_rules is used to mention specific via rules to be used for via insertion -via_occupany_percentage is used to mention the percentage occupancy of via -via_occupancy is used to mention absolute x and y values for via occupany -via_num_cuts is used to mention number of cuts in x and y direction -vias_between_parallel_metals 0/1 is used to control if vias need to inserted between parallel metal layers -parallel_metals_via_spec { <length> <spacing>} is used to mention the length and spacing of vias between parallel metal layers -restrict_via_cuts_overlap is used to limit the via cuts to the overlap area between metals -short_check 0/1 is used to determine if shorts need to be avoided during route creation. Default 1. -min_spacing is used to provide distance in microns at which route mesh creation must stop to avoid short. Default 0.0005 -offset_from_blockages is used to provide a distance in microns at which route mesh creation must stop from blockages in design -offset_from_exclude_scopes is used to provide a distance in microns at which route mesh creation must stop from scopes or blocks excluded in mesh spec. -staples option is used to create staple design and to provide length and spacing of the staples. -hide_spec option is used to hide a spec in GUI under create_route_mesh menu. Default 0 - All specs will be shown in GUI.	db create route_mesh /work/TOP::p -name TOP_m3_POWER -nets { VDD_TOP VDDINT_TOP VSS_GLOBAL } -layer m3 -dir v -width 0.25 -spacing 5.89 -pitch 150 -offsets { 0 61.15 0 0 } -offset_from_blockages 0 -offset_from_exclude_scopes 0 -short_check 0 -min_spacing 0.0005 -drop_via_layers { - m1 } -spec_boundary {select_region {TOP-top_mem_1-top_mem_2-top_mem_3-top_mem_4-top_mem_5-top_mem_6-top_mem_7-top_mem_8-SUBCHIP_INST-PNR_COMPLETE_BLOCK-DSP} } -vias_between_parallel_metals 1 -parallel_metals_via_spec { 0 0 } -via_occupancy_percentage { 100 100 } db edit /work/TOP/rm:BLK_M6 -name BLK_M6 -nets { VDD_SUBCHIP } -layer M5 -dir v -width 25 -spacing 25 -pitch 150 -offsets { 4 51 4 0 } -offset_from_blockages 0 -offset_from_exclude_scopes 0 -short_check 1 -min_spacing 0.0005 -spec_boundary {select_region {+top_mem_3+top_mem_4+top_mem_5+top_mem_6+top_mem_7+top_mem_8} } -vias_between_parallel_metals 1 -parallel_metals_via_spec { 0 0 } -via_occupancy_percentage { 100 100 }

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MESH COMMANDS

db create/edit	stdcell_straps/<std_straps_spec_name> Provide std_straps_spec keyword for db create command. For db edit , provide <std_straps_spec_name> name instead of std_straps_spec keyword. See example on right side for more details.	<parent_cell> -name <name> -layers {<layer1> <layer2> ...} -follow_pins {<pin1> <net1> <pin2> <net2> .. } -offset_from_blockages <val> -offset_from_exclude_scopes <val> -spec_boundary entire core| {bbox { x1 y1 x2 y2 } } |select_region exp { }	Used to create StdCell Straps for follow pin recreation. <parent_cell> mentions the block/cell under which the std cell spec is created.Used only for db create command. Not used in db edit command. -name mentions the spec name used for std cell spec. -layers mentions the layers in which the std cell straps are created. -follow_pins mentions the lef pin and the logical net which is connected to the pin. -offset_from_blockages is used to provide a distance in microns at which std cell mesh creation must stop from blockages in design -offset_from_exclude_scopes is used to provide a distance in microns at which std cell mesh creation must stop from scopes or blocks excluded in mesh spec. -spec_boundary is used to mention areas where route mesh will be created. If 'entire core' option is used , then all the core area will be used by route mesh spec. 'bbox' option can be used to create route mesh only within a rectangular region.'-select_region_exp' is used to provide an expression of scope or design blocks to determine where the grid must be created. Format is -select_region_exp{+block1 -block2 - ... } . '+' means inclusion and '-' means exclusion of a block/region when mesh is created.	db create stdcell_straps /work/pq_li_tx -name RS1 -layers { METAL1 } -offset_from_blockages 0 -offset_from_exclude_scopes 0 -follow_pins { { VSS VSS } { VDD12 VDD12 } } -spec_boundary {select_region {pq_li_1 -mem1 -mem2-R1}} db edit /work/TOP/strap:RS1 -name RS1 -layers { M1 } -offset_from_blockages 1 -offset_from_exclude_scopes 0 -follow_pins { { VDD VDDINT_TOP } { VSS VSS_GLOBAL } } -spec_boundary {}
	switch_mesh/ <switch_mesh_spec_name> Provide switch_mesh_spec keyword for db create command. For db edit , provide <switch_mesh_spec_name> name instead of switch_mesh_spec keyword. See example on right side for more details.	<parent_cell> -name <name> -logical_conn {<pin1> <net1> <pin2> <net2> .. } -master <name> -inst_prefix <name> -spacing {<xval> <yval>} -offsets {<L> <R> <T> <B> } -spec_boundary entire core| {bbox { x1 y1 x2 y2 } } |select_region exp { }	Used to create an array of switches in mesh creation. <parent_cell> mentions the block/cell under which the Switch Mesh Spec is created.Used only for db create command. Not used in db edit command. -name mentions the spec name used for Switch Mesh Spec. -logical_conn mentions the switch pin and netname used for connecting switch pins -master means the switch cell which is used for creating switch array -inst_prefix means the prefix used for creating switches. -spacing means the x and y distance between switches -offsets means the left , right , top and bottom offsets from which switch array creation starts. -spec_boundary is used to mention areas where route mesh will be created. If 'entire core' option is used , then all the core area will be used by route mesh spec. 'bbox' option can be used to create route mesh only within a rectangular region.'-select_region_exp' is used to provide an expression of scope or design blocks to determine where the grid must be created. Format is -select_region_exp{+block1 -block2 - ... } . '+' means inclusion and '-' means exclusion of a block/region when mesh is created.	db create switch_mesh /work/Top -name S1 -master /work/switch_cell -inst_prefix sw_ -spacing { 200 100 } -offsets { 0 0 0 50 } -logical_conn { { VDDC VDD } { VDD VDD_INT } } -spec_boundary {select_region {+ALP}} db edit /work/TOP/sm:S1 -name S1 -master /work/HEAD34_A12TR50 -inst_prefix ins1 -spacing { 100 200 } -offsets { 0 0 0 0 } -logical_conn { { VDD VDD_TOP } { VDDG VDDINT_TOP } } -spec_boundary {entire_core}

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MESH COMMANDS

db create/edit	via_mesh/ <via_mesh_spec_name> Provide via_mesh_spec keyword for db create command. For db edit , provide <via_mesh_spec_name> name instead of via_mesh_spec keyword. See example on right side for more details.	<parent_cell> -name <name> -drop_via_layers {<upper> <lower>} -nets {<net1> <net2> ...} -offsets {<left> >right> <top. <bot>} -spec_boundary entire core| {bbox { x1 y1 x2 y2 } } |select_region exp { } -via_models {<name1> <name2> ..} -via_rules {<name1> <name2> ..} -via_occupancy_percentage {<XVal> <YVal>} -via_occupancy {<XVal> <YVal>} -via_num_cuts {<NX> <NY>} -vias_between_parallel_metals 0|1-parallel_metals_via_spec {<length> <spacing>} -restrict_via_cuts_to_overlap <0|1>-short_check <1|0> -min_spacing <val>	Used to create a spec for route mesh on a specific layer <parent_cell> mentions the block/cell under which the mesh spec is create.Used only for db create command. Not used in db edit command. -name mentions the spec name created for via mesh -drop_via_layers is used to mention upper and lower layers to which vias will be dropped. -nets is used to mention nets for which routes need to be created -offsets is used to mention top , left , bottom and right offsets for starting grid creation. -spec_boundary is used to mention areas where via mesh will be created. If 'entire core' option is used , then all the core area will be used by via mesh spec. 'bbox' option can be used to create via mesh only within a rectangular region. '-select_region_exp' is used to provide an expression of scope or design blocks to determine where the grid must be created. Format is -select_region_exp{+block1 -block2 - ... } . '+' means inclusion and '-' means exclusion of a block/region when mesh is created. -via_models is used to mention specific via models to be used for via insertion -via_rules is used to mention specific via rules to be used for via insertion -via_occupany_percentage is used to mention the percentage occupancy of via -via_occupancy is used to mention absolute x and y values for via occupany -via_num_cuts is used to mention number of cuts in x and y direction -vias_between_parallel_metals 0/1 is used to control if vias need to inserted between parallel metal layers -parallel_metals_via_spec { <length> <spacing>} is used to mention the length and spacing of vias between parallel metal layers -restrict_via_cuts_overlap is used to limit the via cuts to the overlap area between metals -short_check 0/1 is used to determine if shorts need to be avoided during route creation. Default 1. -min_spacing is used to provide distance in microns at which route mesh creation must stop to avoid short. Default 0.0005	db create via_mesh /work/Top -name VM1 -drop_via_layers { METAL6 METAL1 } -offsets { 0.305 1.78 0.305 1.78 } -short_check 1 -min_spacing 0.0005 -nets { } -spec_boundary {entire_core } -vias_between_parallel_metals 1 -parallel_metals_via_spec { 0 0 } -via_occupancy_percentage { 100 100 } db edit /work/TOP/vm:VM1 -name VM1 -drop_via_layers { M3 M4 } -offsets { 0 0 0 0 } -short_check 1 -min_spacing 0.0005 -spec_boundary {entire_core } -vias_between_parallel_metals 1 -parallel_metals_via_spec { 0 0 } -via_occupancy_percentage { 100 100 }

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DRAW AND ROUTE COMMANDS

The below commands are used to perform various manual draw and edit operations

Command	Options	Allowed Arguments	Description	Example
db create	ploc	<parent cell name> -name <name> -layer <name> -loc {<x> <y>} -type <p|g>	Used to insert a power/ground pad in the design <parent_cell> mentions the block/cell under which the pad s created. -name represents the name of the ploc -layer mentions the layer on which the pad is created -loc is used to enter location of new ploc -type p|g mentions if the ploc is power or ground	db create ploc /work/TOP -name ploc4 -layer MET5 -loc {1954.78 1660.12} -type g
	route	<parent_cell> -layers { <layer1> <layer2> ...} -nets {<net1> <net2> ...} -width {<horizontal> <vertical> } -spacing {<horizontal> <vertical> } -via_models {<name1> <name2> ..} -via_rules {<name1> <name2> ..} -via_occupancy_percentage {<XVal> <YVal>} -via_occupancy {<XVal> <YVal>} -via_num_cuts {<NX> <NY>} -vias_between_parallel_metals <0|1> -parallel_metals_via_spec {<length> <spacing>} -restrict_via_cuts_to_overlap <0|1>-points { {<x1> <y1>} {<x2> <y2>} ...}	Used to insert a complex wire route in the design in both horizontal as well as vertical directions. <parent_cell> mentions the block/cell under which the wire is created. -layers is used to mention the horizontal and vertical layers used for route creation -width option is used to mention the width of the wire for horizontal and vertical wires. -spacing option is used to mention the spacing between wires in horizontal and vertical directions -via_models is used to mention specific via models to be used for via insertion -via_rules is used to mention specific via rules to be used for via insertion -via_occupany_percentage is used to mention the percentage occupancy of via -via_occupancy is used to mention absolute x and y values for via occupany -via_num_cuts is used to mention number of cuts in x and y direction -vias_between_parallel_metals 0/1 is used to control if vias need to inserted between parallel metal layers -parallel_metals_via_spec { <length> <spacing>} is used to mention the length and spacing of vias between parallel metal layers -restrict_via_cuts_overlap is used to limit the via cuts to the overlap area between metals	db create route /work/GORDON -nets { VDD_CORE } -layers { M1 } -points { { 1190 4406.44 } { 4567.57 4406.44 } } -width { 0.6 0.6 } -spacing { 0 0 } -vias_between_parallel_metals 1 -parallel_metals_via_spec { 0 0 } -via_occupancy_percentage { 100 100 }
	via	<parent_cell> -name <name> -layers {<upper> <lower>} -loc {<x> <y>} -via_models {<name1> <name2> ..} -via_rules {<name1> <name2> ..} -via_occupancy_percentage {<XVal> <YVal>} -via_occupancy {<XVal> <YVal>} -via_num_cuts {<NX> <NY>} -vias_between_parallel_metals <0|1> -parallel_metals_via_spec {<length> <spacing>} -restrict_via_cuts_to_overlap <0|1>	Used to insert a single via at a point <parent_cell> mentions the block/cell under which the via is created. -layers is used to mention the upper and lower layers used for via insertion -loc is used to mention the x y locations where the via is inserted -via_models is used to mention specific via models to be used for via insertion -via_rules is used to mention specific via rules to be used for via insertion -via_occupany_percentage is used to mention the percentage occupancy of via -via_occupancy is used to mention absolute x and y values for via occupany -via_num_cuts is used to mention number of cuts in x and y direction -vias_between_parallel_metals 0/1 is used to control if vias need to inserted between parallel metal layers -parallel_metals_via_spec { <length> <spacing>} is used to mention the length and spacing of vias between parallel metal layers -restrict_via_cuts_overlap is used to limit the via cuts to the overlap area between metals	db create via /work/TOP -net VDD_TOP -layers { m6 m3 } -loc { 336.276 1656.8 } -vias_between_parallel_metals 1 -parallel_metals_via_spec { 0 0 } -via_occupancy_percentage { 100 100 }

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DRAW AND ROUTE COMMANDS

db delete	{<object_name1> <object_name2>...}	Used to delete mutiple objects in design like wire , via , instance etc	db delete -objs { W-3297-P-1 W-3298-P-1 W-3299-P-1 }
db move	-objs {<object_name1> <object_name2>...} -offset {x y}	Used to move objs from one position to another -objs is used to enter object names -offset is used to enter the x and y offsets to be moved from its current location	db delete -objs { W-3297-P-1 W-3298-P-1 W-3299-P-1 }
db rotate	-objs {<name1> <name2> ..} -dir left|right	Used to rotate objects to left or right by 90 degrees. -objs is used to enter object names -dir is used to enter the direction of rotation either left or right (rotation happens by 90 degrees)	db rotate -objs { /work/TOP/top_mem_8 /work/TOP/top_mem_1 } -dir left
db flip	-objs {<name1> <name2> ..} -dir horizonta|vertical	Used to flip objects either in vertical or horizontal direction -objs is used to enter object names -dir is used to enter the direction of flip either horizontal or vertical	db flip -objs { /work/TOP/SUBCHIP_INST } -dir vertical
db align	objs {<name1> <name2> ..} -dir left|right|top|bottom	Used to align a set of objects on left , right , top and bottom directions -objs is used to enter object names -dir is used to enter the direction in which objects will be aligned (left / right / top / bottom)	db align -objs { /work/TOP/top_mem_2 /work/TOP/top_mem_4 /work/TOP/top_mem_7 } -dir right
db split	wires {<wireid1> <wireid2> ..} -line { {<x1> <y1>} {<x2> <y2>} }	Used to split a set of wires along a horizontal or vertical line -wires is used to provide a set of wire ids -line is used to mention the line along which the line will be split.	db split -wires { W-35-P-1 } -line { { 625.891 1425.78 } { 514.007 1425.78 } }
db join	-wires {<wireid1> <wireid2> ..}	Used to join two wires -wires is used to provide a set of wire Ids to join	db join -wires { W-54-P-1 W-35-P-1 }
db stretch	-objs {<name1> <name2> ..} -edge top|bottom|left|right -length <val>	Used to strecth a set of objects like wires -objs is used to enter object names -edge is used to mention the edge that will be used for stretching -length is used to mention the amount by which the object will be stretch	db stretch -objs { W-33-P-1 } -edge left -length 24.396
db clone	<cell cellname> | <-objs {<object_name1> <object_name2>...} -offset {x y}>	Used to clone a cell or make a copy of object names like wires , instances etc cell <cellname> can be used to make a clone of a GPS session available for a cell in GPS -objs is used to enter object names -offset is used to enter offset values between old and new locations	db clone cell /work/cortexa7core db clone -objs { I-820-P-822 } -offset { 11.217 6.5435 }

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ADVANCED ROUTE AND VIA EDITS

The below commands are used to perform advanced route and via edit operations.

Command	Allowed Arguments	Description	Example
perform adv_route_edit	<cellName> -spec_boundary entire core| {bbox { x1 y1 x2 y2 } } |select_region exp { } -nets <> -type_filter <> -via_rules <> -via_models <> -vias_between_parallel_metals <> -parallel_metals_via_spec <> -via between parallel metals on {<len> <spacing>} -short_check <0/1> -min_spacing <> -via_occupancy_percentage <> -via_occupancy <> -via_num_cuts <> -restrict_via_cuts_to_overlap <0/1> -preview <0/1> -layer <> -selected_wires <> -width_filter <> -width based filter {width|min<width<max} -new_width <new width definition w|w%> -delete_wires <> -fix_edge <> -recreate_vias <>	Used to perform some advanced edit operations on wires. <cellname> mentions the block or instance under which the operation is performed. -spec_boundary is used to mention areas where edit operations will be performed. If 'entire core' option is used , then all the core area will be edited. 'bbox' option can be used to edit routes only within a rectangular region.'-select_region_exp' is used to provide an expression of scope or design blocks to determine where the grid must be edited. Format is -select_region_exp{+block1 -block2 - ... } . '+' means inclusion and '-' means exclusion of a block/region. -nets is used to mention the netnames whose route geometries will be edited. -type_filter is used to mention the route types to edit , 'Manual' option can be used to edit routes created manually, 'Spec' option can be used to edit routes created with spec, 'All' option can be used to edit all types of routes created , -via_rules is used to mention specific via rules to be used for via insertion -via_models is used to mention specific via models to be used for via insertion -vias_between_parallel_metals 0/1 is used to control if vias need to inserted between parallel metal layers -parallel_metals_via_spec { <length> <spacing>} is used to mention the length and spacing of vias between parallel metal layers -short_check 0/1 is used to determine if shorts need to be avoided during route creation. Default 1. -min_spacing is used to provide distance in microns at which route mesh creation must stop to avoid short. Default 0.0005 -via_occupany_percentage is used to mention the percentage occupancy of via -via_occupancy is used to mention absolute x and y values for via occupany -via_num_cuts is used to mention number of cuts in x and y direction -restrict_via_cuts_overlap is used to limit the via cuts to the overlap area between metals -preview 0/1 is used to preview the geometries to be edited -layer is used to mention the layers on which the route edits ar to be done -selected_wires is used to mention the list of wires to be edited. -width_filter {width|min<width<max } specified the minimum and maximum limit of the wire width to be edited -new_width specifies the new width value -delete_wires 0/1 is used to determine if the selected gemoetries ar to be deleted -fix_edge specifies the edge to be fixed for changing route width to the specified value. -recreate_vias 0/1 is used to determine if the vias are to be recreated or not.	perform adv_route_edit /work/TOP -layer M3 -nets {VDDINT_TOP VDD_SUBCHIP } -spec_boundary {entire_core } -type_filter manual -width_filter 1<width<2 -new_width 3 -fix_edge low -vias_between_parallel_metals 1 -parallel_metals_via_spec { 0 0 } -via_occupancy_percentage { 100 100 } -short_check 1 -min_spacing 0.0005

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ADVANCED ROUTE AND VIA EDITS

perform adv_via_edit	<cellname>: -spec_boundary <entire core|bbox {}|select_region exp [exclude_rects{}]> -nets <> -type_filter <> -via_rules <> -via_models <> -vias_between_parallel_metals <> -via between parallel metals <> -parallel_metals_via_spec <> -via between parallel metals on {<len> <spacing>} -short_check <0/1> -min_spacing <> -via_occupancy_percentage <> -via_occupancy arg <> -via_num_cuts arg <> -restrict_via_cuts_to_overlap <0/1> -preview <0/1> -layer <> -selected_vias <> -delete_vias <>	Used to perform some advanced edit operations on vias. <cellname> mentions the block or instance under which the operation is performed. -spec_boundary is used to mention areas where edit operations will be performed. If 'entire core' option is used , then all the core area will be edited. 'bbox' option can be used to edit vias only within a rectangular region.'-select_region_exp' is used to provide an expression of scope or design blocks to determine where the vias must be edited. Format is -select_region_exp{+block1 -block2 - ... } . '+' means inclusion and '-' means exclusion of a block/region. -nets is used to mention the netnames whose via geometries will be edited. -type_filter is used to mention the via types to edit , 'Manual' option can be used to edit vias created manually, 'Spec' option can be used to edit vias created with spec, 'All' option can be used to edit all types of vias created -via_rules is used to mention specific via rules to be used for via insertion -via_models is used to mention specific via models to be used for via insertion -vias_between_parallel_metals 0/1 is used to control if vias need to inserted between parallel metal layers -parallel_metals_via_spec { <length> <spacing>} is used to mention the length and spacing of vias between parallel metal layers -short_check 0/1 is used to determine if shorts need to be avoided during route creation. Default 1. -min_spacing is used to provide distance in microns at which route mesh creation must stop to avoid short. Default 0.0005 -via_occupany_percentage is used to mention the percentage occupancy of via -via_occupancy is used to mention absolute x and y values for via occupany -via_num_cuts is used to mention number of cuts in x and y direction -restrict_via_cuts_overlap is used to limit the via cuts to the overlap area between metals -preview 0/1 is used to preview the geometries to be edited -layer {top bot} is used to mention the via top and bottom layers on which the edits are to be done -selected_vias is used to mention the list of vias to be edited. -delete_vias 0/1 is used to determine if the selected gemoetries are to be deleted	perform adv_via_edit /work/TOP -layer { M4 M5 } -nets {VDDINT_TOP } -spec_boundary {bbox { 2182.07 1623.45 2552.41 2214.23 }} -type_filter manual -via_rules {VIA4_0 } -restrict_via_cuts_to_overlap 1 -vias_between_parallel_metals 1 -parallel_metals_via_spec { 2 3 } -via_occupancy_percentage { 100 100 } -via_num_cuts { 2 3 } -short_check 1 -min_spacing 0.0005

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POWER SPEC COMMANDS

The below commands are used to assign power numbers for Static/Dynamic analysis.

Command	Allowed Arguments	Description	Example
db create rush_current_spec	<parent_cell> -name <name> -spec_boundary entire core|bbox {}|select_region -dynamic_sim_time <value in ns> -dynamic_sim_time_step <value in ps> -domain_pairs { {Ext Net Inst Net} {..} } -user_sta_file <file> -sta_file <file> -switch_turn_on_sequence_file <file> -switch_interval_delay <val> -switch_turn_on_sequence_pattern {<pattern> <start_point>} -cap_cdev { {<net> <esr> <esc> <leakage>} {<net> <esr> <esc> <leakage> ..} -apl_cap_file <file> -cap_per_area <val> -pwl_cap_file <file> -pwl_cap <string>	Used to create rush current spec for Low power analysis. <parent_cell> mentions the parent cell name under which the operation is performed. -name is used to specify the spec name -spec_boundary is used to mention areas where operations will be performed. If 'entire core' option is used , then all the core area will be selected for power assignment. 'bbox' option can be used to assign power only within a rectangular region.'-select_region_exp' is used to provide an expression of scope or design blocks to determine where the power must be assigned. Format is -select_region_exp{+block1 -block2 - ... } . '+' means inclusion and '-' means exclusion of a block/region. -dynamic_sim_time is used to set the 'DYNAMIC_SIMULATION_TIME' -dynamic_sim_time_step is used to set 'DYNAMIC_TIME_STEP' -domain_pairs { {Ext Net Inst Net} {..} } is used to mention the External and Internal Net pairs. -user_sta_file is used to mention the user sta file -sta_file is used to mention the sta file -switch_turn_on_sequence_file is used to mention the file which contains the list of instances provided in switching order -switch_turn_on_sequence_pattern is used to mention the pattern to turn on the switches. -switch_interval_delay is used to specify the switch interval delay -cap_cdev { {<net> <esr> <esc> <leakage>} {<net> <esr> <esc> <leakage> ..} is used to mention the net specific capacitance, resistance and leakage current value. -apl_cap_file is used to mention the apl cap file -'cap_per_area ' is used to provide cap per unit area , this is equally distributed for entire fullchip -pwl_cap_file is used to mention the pwcdev file -pwl_cap {{Arc : <vdd_pin> <vss_pin> } {<Voltage> <esr> <esc> <leakage>} {<Voltage> <esr> <esc> <leakage>}} is used to specify piece wise cap values	db create rush_current_spec /work/TOP -name RPS1 -net_layer_assignment { {VDD_TOP M1} {VDDINT_TOP M4} {VDD_SUBCHIP M3} {VDDINT_SUBCHIP M1} {VSS_GLOBAL M1}} -switch_turn_on_sequence {switch_interval_delay 10 predefined_sequence { A Lower-Left } } -decap { cap_per_area 1 } -spec_boundary {entire_core } -dynamic_sim_time 10 -dynamic_sim_time_step 10 -domain_pairs { {VDDINT_TOP VDD_TOP 1} }

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ANALYSIS COMMANDS

The below commands are used to perform various analysis like Static/Dynamic/CPM/Rampup analysis.

Command	Allowed Arguments	Description	Example
rh perform analysis	<cellName> [-static|-dyamic|-rampup] [-dump_setup_only]	Used to perform analysis. <cellname> mentions the parent cell name under which the operation is performed. -static|-dyamic|-rampup is used to mention the type of analysis to be performed -dump_setup_only if mentioned, dumps the complete setup for the analysis	rh perform analysis /work/TOP -static -dump_setup_only
rh perform pwrcalc	<parent_cell>	Used to create power spec for Static/Dynamic analysis <parent_cell> mentions the parent cell name under which the operation is performed.	rh perform pwrcalc TOP::p

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GUI COMMANDS

These are the commands used for doing GUI operations.

Command	Options	Allowed Arguments	Description	Example
gui zoom	in		Zooms in	gui zoom in
	out		Zooms out	gui zoom out
	top		Zooms to top design	gui zoom top
	prev		Zooms to previous zoom GUI	gui zoom prev
	next		Zooms to next zoom GUI	gui zoom next
	get		returns the bbox for present zoom area	gui zoom get
	rect	<llx> <lly> <urx> <ury>	Zooms to the input bbox area	gui zoom rect { { 1 1} {2 2 }}
	<object_name>		Zooms to the specified object	gui zoom /work/TOP
gui pan	up		Pans Up	gui pan up
	down		Pans Down	gui pan down
	left		Pans Left	gui pan left
	right		Pans Right	gui pan right
gui add	ruler {{<x1> <y1>} {<x2> <y2>}}	-size <int_value>	gui add ruler command adds rulers between the two points that are passed as arguements, size of the ruler can be modified using the switch -size.	gui add ruler {{1 2 } {1 5}} -size 5
		-colour <html_hexcode/colour_name>	The ruler colour can be changed using the option -colour, it supports the hexa decimal html colour codes as well as basic colour names for eg: red/green/blue/orange/yellow/white	gui add ruler {{1 2 } {1 5}} -colour #FFFFFF
	marker {<x> <y>}	-size <int_value>	gui add marker command adds rulers between the two points that are passed as arguements, size of the ruler can be modified using the switch -size.	gui add marker { 1 2 } -size 3
		-colour <html_hexcode/colour_name>	The ruler colour can be changed using the option -colour, it supports the hexa decimal html colour codes as well as basic colour names for eg: red/green/blue/orange/yellow/white	gui add marker { 1 2 } -colour red
		-file <input_file>	Using -file option user can pass the list of coordinates, one location per line, the command will inser markers at all the given locations.	gui add marker -file marker_file
gui select	-bbox {<llx> <lly> urx> <ury>}		Selects objects within the bbox	
	-objs { <object_name1> <object_name2> .... }		Selects and highlights objects that are passed as arguments. objets can include block,regions, route spec etc	gui select -objs { /work/TOP }
	-append		appends the existing selection to highlight the new selection	
gui refresh			Refreshes the GUI	gui refresh
gui dump	<jpg|jpeg>	<output_file_name>	dumps existing gui snapshot in jpg/jpeg format	gui dum jpeg gui.jpeg

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GUI COMMANDS

gui clear	marker		Clears all marker	gui clear marker
	ruler		clear all ruler	gui clear ruler
	all		Clears everything marker, rulers and highlighted objects	gui clear all
	select		clears gui selections	guin clear select
Config set	gui_auto_update	<on/off>	By default GPS GUI is updated after executing the command, this will impact the runtime when executing commands in batch mode. To improve the run runtime user can turn off the GUI updation	config set gui_auto_update off
	gui_lsp	{ {<metal_layer1> <on/off> <on/off>} {<metal_layer2> <on/off> <on/off>} }	Used to turn on/off metal layers. User needs to specify visiblity control as well as selectivity control	config set gui_lsp {{m1 on off}}
		{{all_layers|metal_layers|via_layers|all_insts|all_blockages|misc on off}}	Used to turn on/off visiblity and selectablity of metal layers, vias, instances, blockages etc	config set gui_lsp {{all_layers on off}}
		{{Block|RoutedBlock|HardIP|IO|Switch|Cover|EndCap|PGBump|SignalBump|StdCell|Region on on}}	Used turn on/off visiblity and selectablity of blocks, routed blocks, hardips, IOs, switches, endcaps etc	config set gui_lsp { { Block on on } }
		{{user_placement_blkg|cell_placement_blkg|user_routing_blkg|cell_routing_blkg <on/off> <on/off>}} -layer { <metal_layer1> <metal_layer2>.. }	Used turn on/off visiblity and selectablity of placement and routing blockages. Using -layer option layerwise turn on/off of blockages is possible	config set gui_lsp { {user_placement_blkg on on }} -layer { M1 M2 }
		{{StdCellRow|Ploc off off}}	Used turn on/off visiblity and selectablity for plocs and standard cells	config set gui_lsp {{StdCellRow on on } {Ploc off off}}
		{{InstPin off off}} -layer { <metal_layer1> <metal_layer2>.. }	Used turn on/off visiblity and selectablity of placement and routing blockages. Using -layer option layerwise turn on/off of blockages is possible	config set gui_lsp { { InstPin on off } } -layer { metal1 }
	gui_nsp	{ {net1 <on/off> <on/off>} {net2 <on/off>} }	Used to enable/disable Visiblity and selectablity of power nets	config set gui_nsp {{ VDD on on }}
		{ {all_nets|power_nets|ground_nets <on/off> <on/off>} }		config set gui_nsp {{ all_nets on off }}
	gui_color_scheme	<by_layer|by_net>	Used to change GUI colour scheme. Available colour schemes are to colourise by net or by layer	config set gui_color_scheme by_net
	gui_nightvision	<on/off>	Used to enable/disable nightvision mode	config set gui_nightvision on

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